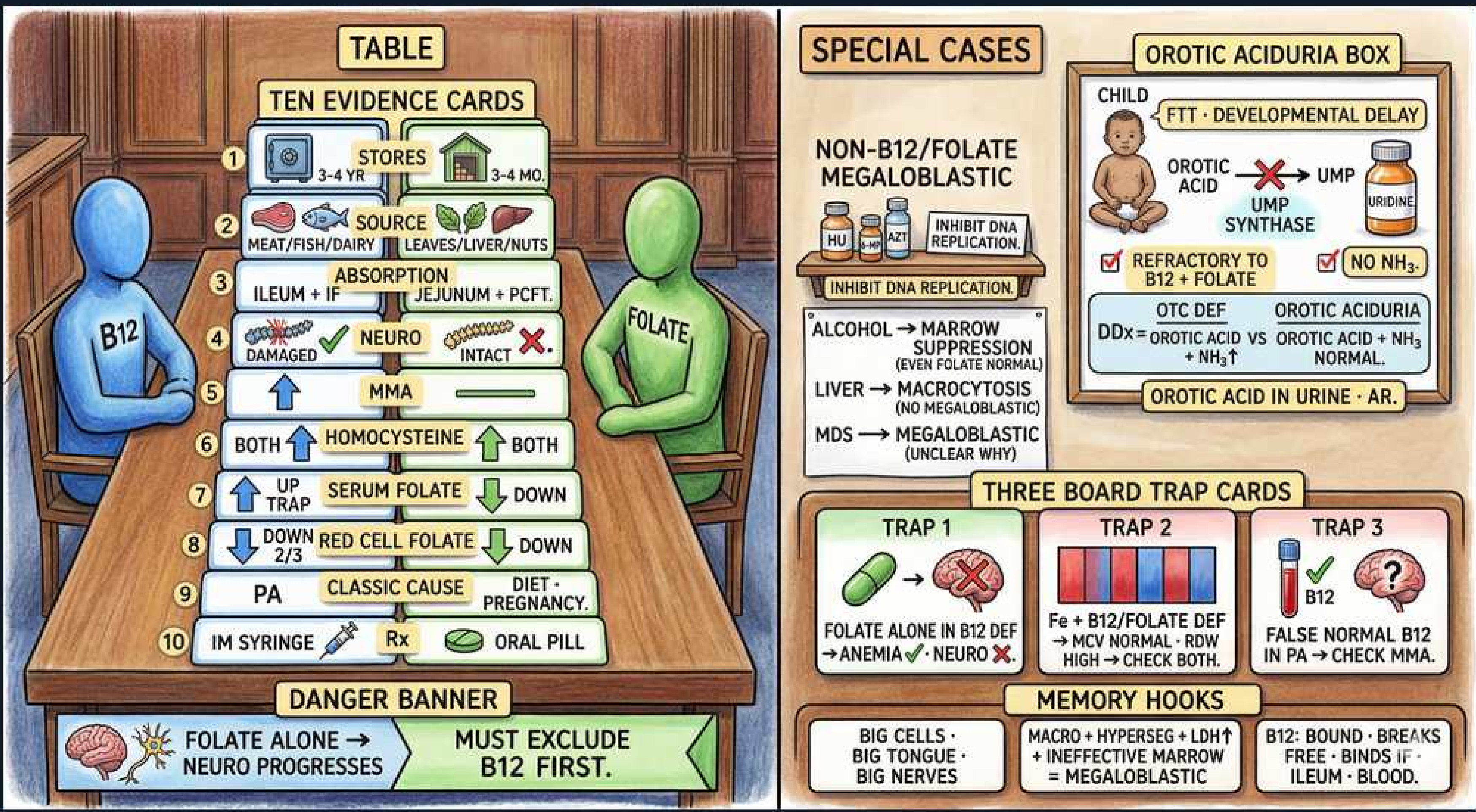
Megaloblastic Anemia: B12 vs Folate



1. B12 vs Folate stores

WHAT YOU SEE

Two players, stores card 3-4 yr vs 3-4 mo

WHAT IT ENCODES

B12 (cobalamin) body stores last 3-4 YEARS; folate stores last only 3-4 MONTHS. This is why folate deficiency develops fast (alcoholics, pregnancy, dialysis) while B12 deficiency takes years to manifest. Board uses the timeline to distinguish etiology.

BOARD TRAP

Picking folate for a slowly progressive macrocytic anemia with neuro signs; the slow course plus neuro points to B12, not folate.

2. Dietary source

WHAT YOU SEE

Meat/fish/dairy vs leaves/liver/nuts

WHAT IT ENCODES

B12 comes ONLY from animal products (meat, fish, eggs, dairy) — strict vegans become deficient. Folate comes from green leafy vegetables, liver, legumes, nuts. Boiling/overcooking destroys folate. Source explains who gets deficient.

BOARD TRAP

Assuming a vegan with macrocytosis has folate deficiency; vegans become B12 deficient because B12 is animal-derived.

3. Absorption site

WHAT YOU SEE

Ileum + IF vs jejunum + PCFT

WHAT IT ENCODES

B12 absorbed in terminal ILEUM bound to intrinsic factor (IF) from gastric parietal cells. Folate absorbed in JEJUNUM via PCFT carrier. Ileal resection/Crohn = B12 deficiency; celiac/tropical sprue (proximal) = folate deficiency.

BOARD TRAP

Terminal ileal Crohn disease or resection causing folate deficiency; ileum is the B12 site, so it causes B12 deficiency.

4. Neuro damage line

WHAT YOU SEE

B12 NEURO damaged vs folate intact

WHAT IT ENCODES

B12 deficiency causes neurologic disease: subacute combined degeneration of dorsal columns/lateral tracts, peripheral neuropathy, dementia. Folate deficiency does NOT cause neuro signs (except neural tube defects in fetus). Neuro findings = B12 until proven otherwise.

BOARD TRAP

Attributing paresthesias and ataxia to folate deficiency; isolated folate deficiency spares the adult nervous system.

5. MMA test

WHAT YOU SEE

MMA elevated only with B12

WHAT IT ENCODES

Methylmalonic acid (MMA) is HIGH in B12 deficiency but NORMAL in folate deficiency — it is the discriminating test. B12 is a cofactor converting methylmalonyl-CoA to succinyl-CoA; without it MMA accumulates. MMA + homocysteine confirm true B12 deficiency.

BOARD TRAP

Using MMA to diagnose folate deficiency; MMA is normal in folate deficiency and only rises in B12 deficiency.

6. Homocysteine

WHAT YOU SEE

Homocysteine up in BOTH

WHAT IT ENCODES

Homocysteine is elevated in BOTH B12 and folate deficiency, so it cannot distinguish them — use MMA for that. Both vitamins are required for the methionine synthase remethylation step. Elevated homocysteine is also a vascular thrombosis risk factor.

BOARD TRAP

Using homocysteine to differentiate B12 from folate; it rises in both, only MMA separates them.

7. Serum vs RBC folate

WHAT YOU SEE

Serum folate vs red cell folate

WHAT IT ENCODES

Serum folate fluctuates with recent intake and can normalize after one meal; RED CELL folate reflects tissue stores over the prior 2-3 months and is the more reliable test. Check both because a normal serum folate can mask true deficiency.

BOARD TRAP

Ruling out folate deficiency on a single normal serum folate; RBC folate is the better marker of tissue stores.

8. Pernicious anemia

WHAT YOU SEE

PA classic cause vs diet/pregnancy

WHAT IT ENCODES

Pernicious anemia (autoimmune anti-IF / anti-parietal cell antibodies, atrophic gastritis) is the classic B12 deficiency cause. Folate deficiency is usually dietary, pregnancy, or drug-induced. PA also raises gastric cancer/carcinoid risk; check anti-IF antibodies.

BOARD TRAP

Confusing PA with folate deficiency; PA is autoimmune loss of intrinsic factor causing B12, not folate, deficiency.

9. Treatment route

WHAT YOU SEE

IM syringe vs oral pill

WHAT IT ENCODES

B12 is replaced IM (or high-dose oral) — IM bypasses the absorption defect in PA. Folate is replaced orally. Treat the right one: giving folate alone in B12 deficiency corrects the anemia but lets neuro disease progress.

BOARD TRAP

Giving oral B12 to a PA patient expecting full correction; IM (or high-dose oral) bypasses the intrinsic-factor defect.

10. Danger: exclude B12 first

WHAT YOU SEE

Banner — folate alone, neuro progresses

WHAT IT ENCODES

NEVER give folate alone before excluding B12 deficiency. Folate corrects the megaloblastic anemia but masks B12 deficiency while irreversible subacute combined degeneration progresses. Always check and replete B12 first or concurrently.

BOARD TRAP

Empirically treating macrocytic anemia with folate; this masks B12 deficiency and allows permanent neurologic damage.

11. Non-B12/folate causes

WHAT YOU SEE

Box — drugs inhibit DNA replication

WHAT IT ENCODES

Megaloblastic changes without B12/folate deficiency: drugs that inhibit DNA synthesis — methotrexate, hydroxyurea, azathioprine, 5-FU, trimethoprim, zidovudine. Hereditary orotic aciduria too. Megaloblastosis = impaired DNA synthesis, not always vitamin deficiency.

BOARD TRAP

Assuming every megaloblastic smear means B12/folate deficiency; antifolate and antimetabolite drugs cause it with normal vitamin levels.

12. Orotic aciduria

WHAT YOU SEE

Child, FTT, UMP synthase, orotic acid in urine

WHAT IT ENCODES

Hereditary orotic aciduria: UMP synthase defect causes megaloblastic anemia REFRACTORY to B12/folate, failure to thrive, developmental delay, and HIGH orotic acid in urine with NO hyperammonemia. Treat with uridine. Distinguish from OTC deficiency (high orotic acid WITH hyperammonemia).

BOARD TRAP

Confusing orotic aciduria with OTC deficiency; orotic aciduria has no hyperammonemia, OTC deficiency does.

13. Trap 1: folate alone in B12 def

WHAT YOU SEE

Trap card — anemia up, neuro down

WHAT IT ENCODES

Folate given alone in B12 deficiency improves the anemia but worsens/unmasks neurologic disease. The hematologic response falsely reassures while myelin damage continues. This is the highest-yield megaloblastic board trap.

BOARD TRAP

Reading an improving CBC as cure; neuro disease silently progresses if B12 was the real deficiency.

14. Trap 2: combined Fe + B12/folate

WHAT YOU SEE

Trap card — MCV normal, RDW high

WHAT IT ENCODES

Concurrent iron deficiency (microcytic) plus B12/folate deficiency (macrocytic) can yield a NORMAL MCV with a HIGH RDW (dimorphic smear). A normal MCV does not exclude megaloblastic anemia — check the smear and RDW.

BOARD TRAP

Excluding megaloblastic anemia because MCV is normal; mixed deficiency normalizes MCV but raises RDW.

15. Trap 3: false-normal B12

WHAT YOU SEE

Trap card — B12 normal in PA, check MMA

WHAT IT ENCODES

Serum B12 can be falsely normal in true deficiency (assay limitations, high transcobalamin). If clinical suspicion is high, check MMA and homocysteine, which rise in true B12 deficiency. Do not rely on a borderline-normal B12 alone.

BOARD TRAP

Ruling out B12 deficiency on a normal serum B12; confirm with MMA when suspicion is high.

16. Memory hooks

WHAT YOU SEE

Big macro / big tongue / big nerves

WHAT IT ENCODES

Megaloblastic hallmarks: macrocytosis with hypersegmented neutrophils, ineffective intramedullary hemopoiesis (high LDH, high indirect bilirubin), glossitis (smooth red tongue). B12 also gives neuropathy. These features anchor the diagnosis on the smear and exam.

BOARD TRAP

Missing hypersegmented neutrophils as the smear clue; they are the classic megaloblastic marker distinguishing it from other macrocytosis.